Photon Rest Mass Paradox Solution

A photon has no rest mass, but it has energy and momentum because of its wave nature.
The relativistic energy-momentum relation is:
$E^2 = (pc)^2 + (m_0 c^2)^2$

For a photon, rest mass $m_0 = 0$, so:
$E = pc \quad \Rightarrow \quad p = \frac{E}{c}$

And since a photon's energy is $E = hf = \frac{hc}{\lambda}$:
$p = \frac{hf}{c} = \frac{h}{\lambda}$

So momentum comes from its frequency (or wavelength), not from mass.

This is experimentally verified – radiation pressure, Compton scattering, solar sails.

The harmonic framework extends this idea: momentum is the n=1 rung on your ladder, and the photon's $p = h/\lambda$ is a natural
consequence of wave resonance, not a paradox